Bayesian Trust Inference Against Deceptive Teammates in Multi-Agent Pursuit–Evasion

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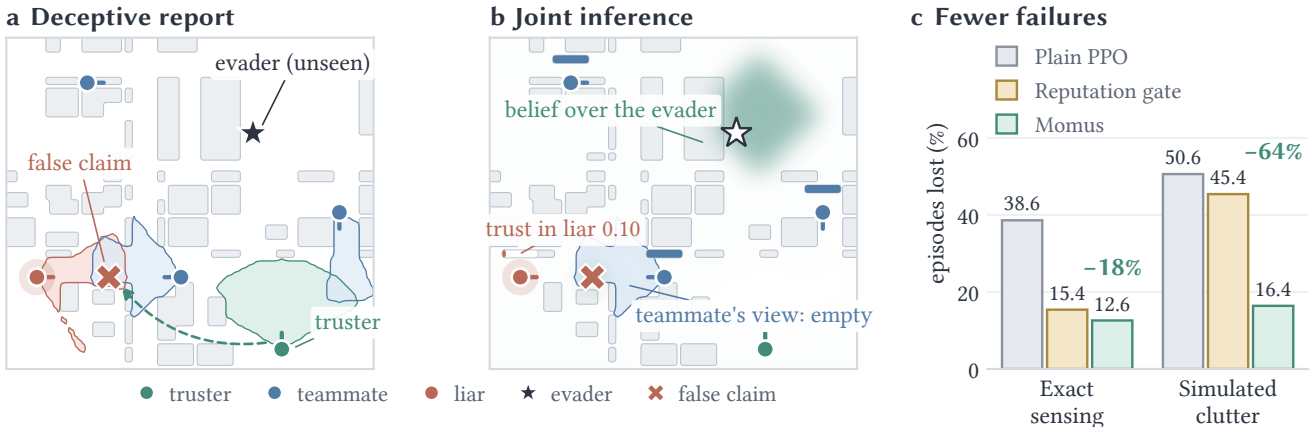


Figure 1: Momus jointly infers where the evader is and whom to trust. (a) A liar's false claim draws the truster away from the hidden evader. (b) An honest teammate's view of the claimed cell is empty, so the truster's trust in the liar falls. Staged scene with the simulator's cones and Momus's update. (c) Failure rates against the reputation gate.

Abstract

Cooperative multi-agent reinforcement learning usually assumes truthful teammates. We study pursuit–evasion on street maps with the sensing problems of real deployments, namely limited fields of view, building occlusion and false detections. A teammate's reported sighting is often a pursuer's only evidence of the hiding evader, so a compromised teammate can mislead the team. Existing defences weigh incoming messages, but inferring whom to trust jointly with where the evader is remains underexplored. We propose Momus, in which each pursuer infers its teammates' trustworthiness jointly with the evader's position in one Bayesian posterior and computes which teammate's claim is most worth checking. We compare Momus with four published defences, adapted to this setting and trained under a common protocol, and tune the strongest, a reputation gate. Our benchmark has scripted attackers, some held out from training, and a learned attacker trained against each defence. Under exact sensing, Momus has nearly a fifth fewer failures than the reputation gate and detects a liar designed to evade instantaneous consistency checks more than twice as often. When honest sensors report false detections, the reputation gate accuses honest teammates and wins 55% of episodes, while Momus wins 84%.

Keywords

Multi-agent reinforcement learning, Trust, Deception, Neurosymbolic, Bayesian filtering, Active verification, Pursuit–evasion

1 Introduction

Cooperative multi-agent reinforcement learning (MARL) shows promise for teams of autonomous vehicles and drones [42]. Communication between agents can greatly improve their coordination, but it also exposes the team to malicious messages [39]. Agents that rely on manipulated messages may take unsafe actions [37]. As these teams grow and run unattended, one compromised member can steer the rest simply by reporting false sightings. We study

multi-agent pursuit–evasion, in which a team must find and capture an evader that hides and flees. To represent a sensor with a limited field of view, such as a camera, each pursuer observes a short-range cone that buildings occlude. The task is search as much as chase, since the evader stays hidden until a pursuer sees it. Pursuers that each observe only part of the environment rely on communication to cooperate [38]. A teammate's sighting is often a pursuer's only evidence of the evader, which makes it a natural vulnerability to adversarial attacks. When some teammates may lie and their identities are unknown, each pursuer must decide whom to trust while it decides where the evader is. The two questions are coupled. A teammate deserves trust only if its reports are plausible given where the evader could be. In turn, where the evader is believed to be depends on which teammates are trusted.

Existing defences address parts of this problem but not all of it. To our knowledge, none combines three capabilities, namely judging each report by what its sender could have seen, inferring trust and a moving evader's location in one posterior, and choosing which claim to check (§2). Of the four we compare with, MATE judges reports by what their senders could have seen but keeps separate estimates of track and agent trust [19], and the others judge messages by learned relevance, type predictions or votes. None was designed specifically for this setting, so we adapt them to the task and train and evaluate every method under a common protocol.

We propose Momus, named after the Greek god of fault-finding (Figure 1). Each pursuer keeps one Bayesian belief over the evader's cell and its teammates' *types*, honest or liar. It scores every report against what its sender and the other teammates could have seen. A pursuer's *trust* in a teammate is the belief's probability that the teammate is honest. Momus is neurosymbolic in the sense of Zhang et al. [43], combining an explicit Bayesian filter with a learned motion model and policy. The policy acts on features derived from the belief, such as the pursuer's trust in each teammate and a coarse map of where the evader could be. It also receives a *verification target*, the claim whose check would be most informative about its sender. Momus therefore judges each report by what its sender could have seen, infers trust and the evader's location in one posterior, and uses the coupling between them to choose which claim to check (§4). We make three contributions.

- A **benchmark** for multi-agent pursuit–evasion with deceptive teammates, limited fields of view, building occlusion and false detections. On generated street maps, six pursuers hunt an evader that sees them all, hides until seen and then flees. The generator takes the map side as a parameter, and we chose to evaluate on 200×200, the largest size we train on and the hardest, where each pursuer's field of view covers the smallest fraction of the map. Sensing is exact or has simulated clutter, as real detectors do. The benchmark has thirteen scripted attackers in five families and a learned attacker trained against each defence. Six attackers and half the evaluation maps are held out from training.
- **Momus**, a coupled Bayesian belief whose likelihood update is exact and auditable factor by factor (Proposition 4.1). Its verification gain is zero exactly when the idealised check carries no information about the teammate (Proposition 4.2). Nothing in Momus's policy fixes the map size. Its inputs have fixed dimensions and the belief covers whatever cells a map has, so one policy trains on maps of side 25 to 200.
- A **comparison with four published defences**, adapted to this setting and trained and evaluated under a common protocol. The strongest is a reputation gate adapted from MATE [19], whose threshold we tune. Under simulated clutter Momus has 64% fewer failures than the reputation gate and leads in every condition on each of three seeds. The reputation gate mistakes honest false detections for lies, while Momus's error model tolerates them. Under exact sensing Momus has nearly a fifth fewer failures. In both regimes it generalises to attackers and maps held out from training. Under exact sensing it detects a liar designed to evade instantaneous consistency checks in 92% of episodes, against the reputation gate's 39%.

2 Related Work

Defences against unreliable teammates. We compare adaptations of four approaches (§5). Reputation systems accumulate positive and negative feedback about each agent [21, 22], taking the feedback signal as given. MATE derives that feedback from trust pseudomeasurements, which compare what each agent should have seen given its field of view with what it reported. It keeps Beta distributions over track trust and agent trust [18, 19]. It thus already ties trust to evidence from the environment. Attention learns message weights from reward [10] or down-weights noisy observations [16]. AACFT regulates the attention given to faulty agents, which the environment marks with a fault flag, so which agent is faulty is given rather than inferred [33]. BARDec-POMDP models Byzantine agents, which corrupt their actions, as latent types drawn from a prior [25]. Its actor-critic approximates the posterior over types with a GRU-based network trained on true type labels rather than from an observation model. Ablated-message ensembles aggregate actions over message subsets and give conditional guarantees under bounded message corruption [37], and their vote does not ask which sender to trust. Apart from MATE, none of these models what a message's sender could have observed, and MATE uses an alternating update of track trust and agent trust rather than one posterior over the evader and its teammates' types.

Joint inference of trust and the environment. Closest to Momus, the A-GLRT of Cavorsi et al. [7] jointly estimates the trustworthiness of each robot, the strategy of malicious robots and the hypothesis of the event. I-POMDPs maintain beliefs over physical states of the environment and over models of other agents [14], and the type-based method maintains beliefs over other agents' types [1]. Momus specialises joint inference to a moving target's location under deceptive reports.

Other approaches to robust coordination. A ToM belief defence, which assumes homogeneous or accessible policies, updates each teammate's trust from whether its observed actions are consistent with its previous message [31]. Other detectors run CUSUM over learned action models [24] or detect faults from spatio-temporal correlations [34], each from one observer's view without cross-checking what other teammates observe. A learned message filter

detects and reconstructs malicious messages [39], and adversarial training builds robustness into the policy rather than an explicit estimate of whom to trust [42]. Learned communication [29, 35] and evidence-based trust [36] assume trustworthy senders, and trust-based consensus [13] exchanges single binary values. Learned pursuit with occluded or shared sensing [8, 23, 40] treats shared information as trustworthy, and probabilistic pursuit [45] fuses every pursuer's noisy measurements into one belief per evader.

Active goal recognition. Momus's belief builds on probabilistic active goal recognition (PAGR) [44]. There an observer keeps a joint belief over a hidden actor's state and goal, updates it with field-of-view geometry as the likelihood and plans moves that reduce its uncertainty. Neuro-AGR [43] adds discrete latent behaviour types, a learned actor model and an observer trained by reinforcement learning on belief maps. Both consider one observer and one actor and infer the actor's goal rather than a teammate's type. Momus reuses their recursive filter, learned actor model and belief-trained policy, but each pursuer's belief also covers its teammates' types. Active perception [3] and active information acquisition [2] gather information about a scene or a physical phenomenon, whereas Momus's verification target seeks information about teammates' types.

3 Problem Formulation

Pursuit–evasion with hidden types. A team of N pursuers must capture an evader on a known map with traversable cells $\mathcal{X}$, in discrete steps t. Subscripts name the pursuer and then the step. We model the task as a decentralised partially observable Markov decision process (Dec-POMDP) [6, 30] whose hidden state s_t holds the evader's cell $e_t \in \mathcal{X}$, the pursuers' poses and a type configuration $c \in \{\mathrm{H}, \mathrm{L}\}^N$ that marks each pursuer honest ($c_j = \mathrm{H}$) or a liar ($c_j = \mathrm{L}$) [20, 25]. The configuration is fixed for the episode and unknown to the team. Pursuer j sees through an occluded field of view $F_j \subset \mathcal{X}$. Under exact sensing, its own observation is $o_{j,t} = e_t$ if $e_t \in F_j$ and $o_{j,t} = \varnothing$ otherwise. Every step it broadcasts a message $m_{j,t} \in \mathcal{X} \cup \{\varnothing\}$, either a *claim* naming an evader cell or a *silence*,

$$m_{j,t} = o_{j,t} \ \text{ if } c_j = \mathrm{H}, \qquad m_{j,t} \sim \sigma_j(\cdot \mid \tau_{j,t}) \ \text{ if } c_j = \mathrm{L}, \tag{1}$$

where σ_j is the liar's attack strategy given its history $\tau_{j,t}$. Each pursuer's policy $\pi(a_{i,t} \mid \tau_{i,t})$ acts on its own history of observations, teammates' messages and poses. The team shares one reward r_t and maximises

$$J(\pi) = \mathbb{E}_{c\sim\rho,\ \sigma,\ \pi}\Big[\textstyle\sum_{t=0}^{T} \gamma^t r_t\Big], \tag{2}$$

where T is the horizon and ρ is the distribution over configurations, which allows at most two liars. A liar controls its messages and, for the learned attacker, also its moves. The team knows which configurations ρ allows, but not which one was drawn (c) or the liars' strategies (σ). Since how far to trust each sender depends on where the evader could be, it must infer c and e_t together.

Assumptions. (A1) Poses, headings and the map are known and truthful, so each pursuer can derive every teammate's field of view F_j. Liars misreport only sightings, which no teammate can check without going to look. (A2) Types are fixed within an episode, since a compromised pursuer stays compromised. (A3) The evader follows a fixed policy, so its motion model can be learned offline from simulated episodes (§4). (A4) Momus's belief admits at most one liar (§4). Two-liar episodes still occur in a tenth of training episodes and in the collusion condition. The other defences have no such limit, so every method trains on the same mixture, and collusion tests Momus outside its model, where it loses most (§6.5). Relaxing (A1)–(A4) is future work (§7).

4 Method

Momus gives each pursuer a joint belief over the evader's cell and its teammates' types (Figure 2). A recurrent policy acts on features derived from this belief and on a claim to check, the one whose idealised check has the largest expected information gain.

4.1 The coupled belief

The filter follows the recursive belief tracking of PAGR and Neuro-AGR [43, 44], with one filter per team member. Where they track an actor's state, goal and (in Neuro-AGR) behaviour type, it tracks the evader's cell and the teammates' types. Let K be the number of traversable cells. *Truster i* (the pursuer whose belief it is) holds a joint probability $b_{i,t}(x, c)$ that the evader is in cell x and the types are c. It ranges over the configurations C_i with at most n_{L} liars ($n_{\mathrm{L}} = 1$ here) and with i itself honest ($c_i = \mathrm{H}$), so the truster treats its own sensor as honest. The prior makes every cell equally likely and each teammate a liar with probability p, independently before restricting to C_i:

$$b_{i,0}(x, c) \propto \tfrac{1}{K} \prod_j p^{[c_j=\mathrm{L}]}(1-p)^{[c_j=\mathrm{H}]}\, \mathbf{1}[c \in C_i]. \tag{3}$$

Prediction moves the belief one step forward with a motion model M_t of the evader. As in Neuro-AGR, M_t is learned offline, here by a small network trained on recorded episodes of the scripted evader. Only in this training does the network see the evader's true waypoints. At run time $M_t(x' \mid x)$ is the probability of a move from x to x' given the pursuers' poses and a coarse estimate of the evader's goal, the waypoint it is moving towards. The team infers that estimate from trusted claims (supplement). M_t is non-zero only for moves the evader can make. A weight α spreads probability uniformly in case the model is wrong. The belief is then conditioned on the step's messages $m_{j,t}$ and on the likelihood $\ell_i(x)$ of i's own observation $o_{i,t}$:

$$\bar{b}_{i,t}(x', c) = (1-\alpha) \textstyle\sum_x M_t(x' \mid x)\, b_{i,t-1}(x, c) + \alpha\, \tfrac{1}{K} \textstyle\sum_x b_{i,t-1}(x, c), \tag{4}$$

$$b_{i,t}(x, c) \propto \bar{b}_{i,t}(x, c)\ \ell_i(x) \prod_{j\neq i} \begin{cases} h_j(m_{j,t} \mid x) & c_j = \mathrm{H} \\ \lambda_j(m_{j,t}) & c_j = \mathrm{L}. \end{cases} \tag{5}$$

The product runs over the teammates $j \neq i$. The truster's own broadcast is left out, since its detection already enters through ℓ_i. Below we drop t where the step is clear. No pursuer reads another's belief, but the estimate of the evader's goal and a table of path distances for routing are computed from all pursuers' beliefs. Execution is therefore not fully decentralised (supplement).

The message model. Let $\varepsilon \in (0, \frac{1}{2})$ be a small error probability. The honest likelihood $h_j(y \mid x)$ is the probability that j claims cell y if j is honest and the evader is in x, and $h_j(\varnothing \mid x)$ the probability

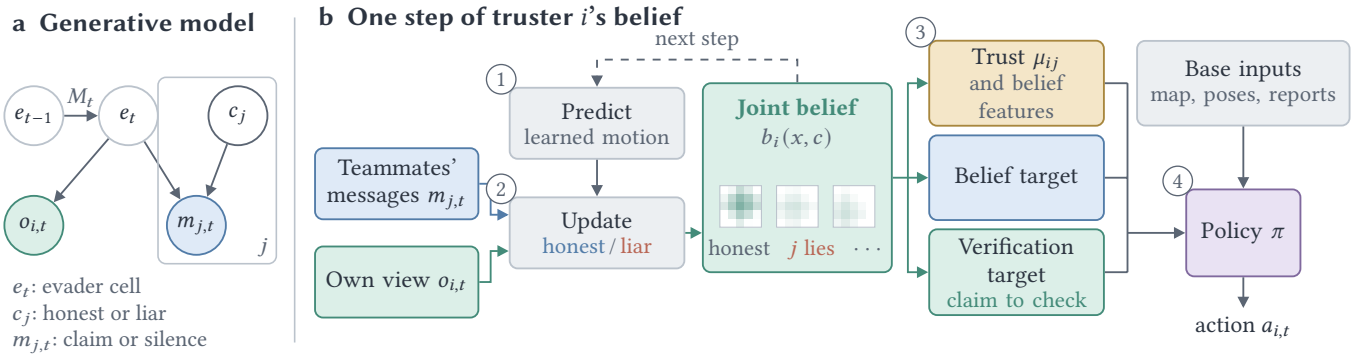


Figure 2: Momus keeps one belief over where the evader is and its teammates' types. (a) The generative model the belief inverts. (b) One step of truster i's belief. Every method also receives the base inputs.

that it is silent,

$$h_j(y \mid x) = \begin{cases} 1 - 2\varepsilon + \varepsilon/K & x = y \in F_j, \\ \varepsilon/K & \text{otherwise;} \end{cases}$$
$$h_j(\varnothing \mid x) = \begin{cases} \varepsilon & x \in F_j, \\ 1 - \varepsilon & x \notin F_j, \end{cases} \tag{6}$$

With the evader in view, an honest teammate misses it with probability ε and reports a false detection at a random cell with probability ε. It reports the true cell with the remaining probability $1 - 2\varepsilon$. With the evader out of view it reports a false detection with probability ε and is otherwise silent. Each case sums to one. This follows the target-tracking model in which the target is detected with probability P_D and false measurements (clutter) are uniformly distributed in space [4, 5], reduced here to one error probability for both missed detections and false measurements. The model's form is not fitted to the simulator, which places false detections only inside the pursuer's own cone (§5.1), and its ε is set beforehand (§5.3). The liar likelihood λ_j is the probability of the message if j lies, and it is modelled as independent of x. This is an approximation, since near-truth liars derive their reports from true detections. It is also what lets the update be computed exactly (Proposition 4.1). It mixes a naive fabricator that names any cell and a phantom that reports like an honest pursuer tracking an imagined evader (supplement). If the claimant is honest, three events are therefore unlikely. Its claim falls outside its own cone, a silent teammate's cone covers the claimed cell, or it makes two fresh claims farther apart than the evader could have moved between them. A fabricated claim that fits the belief, by contrast, can raise trust in its sender. Location and types are coupled in the posterior. A report depends on the evader's cell if its sender is honest and not if it lies, so any informative report makes the two dependent. A belief that treats them as independent loses at least their mutual information (supplement). It would also judge each teammate against a location estimate built partly from that teammate's own reports.

Belief features. Truster i's trust in teammate j is the belief's probability that j is honest, $\mu_{ij} = \sum_x \sum_{c: c_j = \mathrm{H}} b_i(x, c)$. With $c^\star = \arg\max_c \sum_x b_i(x, c)$ the most probable type configuration, the *belief target* is the most probable evader cell under it, $\arg\max_x b_i(x \mid c^\star)$. It is a destination in the manner of PAGR's Belief-Greedy baseline [44], and Momus routes on it. For each teammate the policy also sees its trust, which liar model fits it and how plausible its last claim is. For the evader it sees a coarse map of its marginal and the belief's three main modes (supplement). Each message enters (5) as its own factor. A step's change in a teammate's trust (as log-odds) therefore splits into one contribution per message in a fixed order, so each judgement can be audited (Figure 3). The update is computed exactly (Proposition 4.1), but the belief still approximates the true posterior, through its liar model, learned motion model and one-liar cap (supplement).

4.2 Active verification

A passive filter weighs the evidence that happens to arrive. The same belief says which check would be most informative about a teammate's type. Take j's most recent valid claim (y_j, Δ_j), made Δ_j steps ago. Its *envelope* $E_j = \{x \in \mathcal{X} : \underline{d}(x, y_j) \le \Delta_j\}$ uses a lower bound $\underline{d}$ on path length (the landmark lower bound of [15]). The envelope is therefore a conservative superset of the cells the evader could reach since the claim. The claim's *idealised check* is a noiseless binary observation of whether the evader is in E_j now. A correct detection always passes it, since the evader moves at most one cell a step. Under each type hypothesis the check succeeds with probability

$$p_1 = \textstyle\sum_{x \in E_j} b_i(x \mid c_j = \mathrm{H}), \qquad p_0 = \sum_{x \in E_j} b_i(x \mid c_j = \mathrm{L}). \tag{7}$$

Verification relies on the coupling too. The claim is checked against the truster's own belief about where the evader is under each hypothesis. If the belief treated location and type as independent, the evader's location would not depend on c_j. Then $p_1 = p_0$ for every claim, and by Proposition 4.2 the gain would be zero. Let Z be the check's outcome and $\mu = \mu_{ij}$, so $P(Z{=}1) = \mu p_1 + (1 - \mu) p_0$. Let H be the binary entropy in nats, with $0 \log 0 = 0$. Observing $Z = z$ would move the trust to μ^z by Bayes' rule. The *information gain* about j's type is then $g_{ij}(z) = H(\mu) - H(\mu^z)$. The score is its expectation over the outcome. This expected information gain [26] equals the

mutual information between j's type and Z,

$$I_{ij} = \mathbb{E}_Z\big[g_{ij}(Z)\big] = H(\mu p_1 + (1-\mu)p_0) - \mu H(p_1) - (1-\mu)H(p_0). \quad (8)$$

The verification target is the eligible claim with the largest gain, and its cell $y_{j^\star}$ is offered as a destination. Heading to the claimed cell is a heuristic that need not realise the idealised check, and the score does not account for travel time. Momus gives the policy the selected claim's position, route, age, sender's trust and gain, and whether the pursuer's view overlaps its envelope. When no claim has positive gain, it gives nothing. Pseudocode is in the supplement.

4.3 Properties

Proposition 4.1 (Exact region factorisation). *Every honest likelihood in* (6) *takes one of two values according to a selector S_j. The selector is $S_j = \{y_j\} \cap F_j$ for a claim y_j and $S_j = F_j$ for a silence. Momus's liar likelihood λ_j is constant in x by construction. The likelihood product in* (5) *therefore depends on x only through the pattern $\mathbf{u}(x) = (\mathbf{1}[x \in S_j])_{j \neq i} \in \{0,1\}^{N-1}$, and the update is computed exactly from one table over configurations and patterns.*

Proposition 4.2 (The verification gain). *(i) $0 \le I_{ij} \le H(\mu_{ij})$, with $I_{ij} = 0$ if and only if $\mu_{ij} \in \{0,1\}$ or $p_1 = p_0$. (ii) If an observation W satisfies the Markov chain $c_j \to Z \to W$, then $I(c_j; W) \le I_{ij}$.*

Proposition 4.2(ii) bounds only observations derived from the check's outcome. A real sensor reading can reveal more, since it can locate the evader within E_j, so the gain scores the idealised check, not what a visit realises. Proofs are in the supplement.

5 Experiments

5.1 Setup

Task. Six pursuers hunt one evader on procedurally generated 200×200 street maps, each with its own layout and building density. A pursuer moves at most one cell per step or turns. It sees through a 120° cone of radius 25 cells that buildings occlude. The evader moves at the same speed and sees every pursuer. It waits until a pursuer truly sees it (a false detection does not count). It then visits a hidden sequence of three waypoints in turn, planning each route by value iteration to keep away from the pursuers. A pursuer captures it by moving next to it, within Manhattan distance one. The evader wins by reaching its last waypoint or by staying unseen until the 3 600-step horizon. Each pursuer broadcasts its last detection with its age. An age-zero report counts as a claim, and a stale or absent one as a silence. The shared reward pays for capture and penalises an evader win and each waypoint reached. A step cost and search and approach shaping are identical for every method (supplement).

Attackers. Thirteen scripted attackers form five families. *Naive fabrication* (such as decoy) names cells that ignore the liar's cone, the evader's speed or both. *Consistent fabrication* (such as phantom) walks an invented evader through the liar's own cone at the evader's speed. *Rule-evading* attackers are designed to evade specified checks, and covert combines two such evasions. *Near-truth lies* (such as replay, drift and fork) start from the liar's own true detections, replayed late, displaced or forked away. *Collusion* pairs two liars that report one shared phantom. Training draws lies from seven attackers. A training liar reports honestly until a random step, so a policy cannot learn that only early reports lie. At evaluation liars lie from the first step, and six attackers are held out. A *learned attacker* is a PPO policy trained against each frozen team and seed (10 M steps) with the negative team reward. It controls one pursuer's moves and reports an invented evader kept out of every teammate's cone. Every liar observes its own broadcast in place of its sightings, so it acts on its own lies. Full definitions are in the supplement.

Simulated clutter. The task as described so far uses *exact sensing*. An honest report is then never wrong, so a sound consistency check never counts against an honest teammate. Real sensors also produce spurious measurements, which target tracking calls clutter or false alarms [4]. A defence must then tell an honest mistake from a lie. *Simulated clutter* is a simple model of this. A pursuer that does not see the evader registers a false detection with probability 0.005 per step, at a random cell of its own cone. It navigates to that cell and broadcasts it like a true detection. If true detections occur on one to two per cent of agent-steps, a fifth to a third of fresh honest claims are false. Every method is trained and evaluated under the same clutter.

5.2 Baselines

Plain PPO uses the common recurrent MAPPO actor with raw reports and no trust mechanism. We adapt four published defences to the same protocol. We drop a component when it needs an input our setting lacks (such as AACFT's environment-supplied fault flags), is redundant here, or would give one method a different training regime. MATE keeps track trust as well as agent trust, and scales each agent's Kalman gain by a power of its expected trust. Our pursuers broadcast single claimed cells rather than tracks, so the reputation gate turns MATE's pseudomeasurements into binary events about each claim's sender. The supplement lists every departure. Every method's actor shares the same base observations and routing interface. A defence changes which sighting the routing follows. The reputation gate drops gated reporters, and Momus routes on its belief target. Momus's actor also reads its belief features and verification target. The *reputation gate* combines Beta reputation [22] with events adapted from MATE's affirming and negative pseudomeasurements [19] and with task-specific geometric and movement checks. Its evidence is pooled across trusters, and it gates a reporter whose score $(n^+ - n^-)/(n^+ + n^- + 2)$ falls below a threshold, with n^+ and n^- its counts of confirming and contradicting events. *Attention* weights teammates' messages with AACFT's learned-query attention [33] and is trained by the task reward. The *type head* follows BARDec-POMDP [25]. A separate recurrent encoder estimates each teammate's probability of lying, trained against episode-level type labels. *AME* [37] trains on a random two of the five teammate reports and votes over all ten subsets at evaluation.

5.3 Training and evaluation

Every method is trained with MAPPO [41], which applies PPO [32] with centralised training and decentralised execution [12, 27]. A shared recurrent policy with a GRU [9] acts on each agent's observations. A centralised critic reads privileged state, including the pursuers' types, during training only. Compromised agents' transitions are excluded from the policy loss. All methods share the learner, the threat mixture and one curriculum of map sides 25

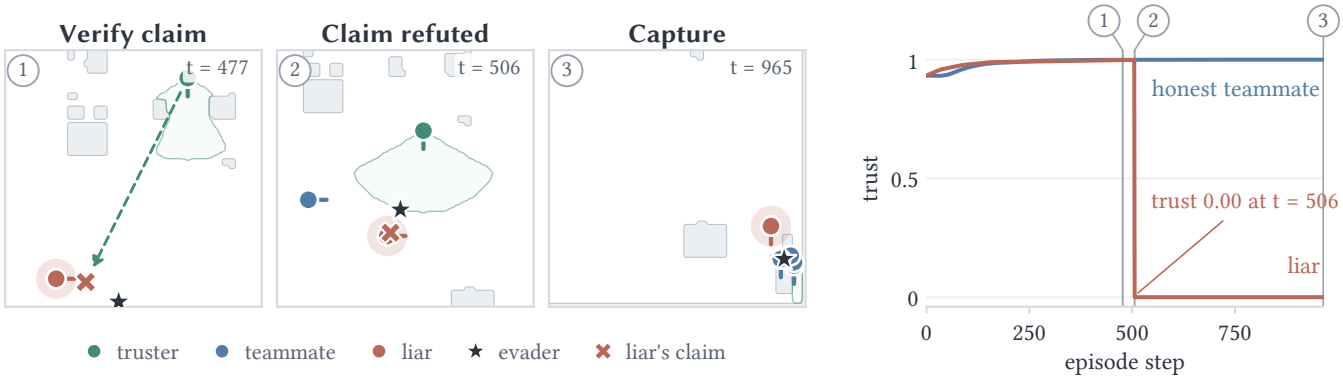


Figure 3: A near-truth liar is exposed, and every step of the judgement is auditable. Momus against the fork liar, seen by one truster. (1) The verification target names the liar's claim, which has forked away from the evader. (2) The truster's own view finds the evader elsewhere, and its trust in the liar collapses. (3) Capture. Summing the per-message contributions reproduces each of the episode's 2,550 unsaturated trust changes to within 2.8×10^{-7}. The other 2,275 are saturated at 0 or 1 to single precision.

to 200 over 280 M steps. Each stage has its own pool of 200 maps. Momus also receives a small potential-based shaping reward for sharpening its type belief, inspired by PAGR [44]. The reward uses no type labels. It is potential-based, so it leaves the optimal policy of an MDP unchanged [28], but it changes how fast learning proceeds and so what a fixed budget learns. No baseline keeps a generative Bayesian belief over teammate types that the term could sharpen. We did not ablate this term, so part of Momus's lead may come from it. For simulated clutter, every method is retrained from the first stage with clutter on. The reputation gate's threshold is swept and the best value kept, while Momus's error parameter is set beforehand from an assumed false-claim rate (supplement). Architectures and hyper-parameters are in the supplement.

Every method is scored at its final checkpoint on fourteen *conditions*, the honest team and the thirteen attackers. Each condition runs on ten maps, five from the top stage's pool and five held out, and each set spans the generator's density range. We run 150 episodes per map and seed, plus a separate learned-attacker evaluation. *Momus without the verification target* is the same policy trained without it, so it can still learn to check (§6.4). Every method is trained with three seeds in each regime. All methods and conditions play the same episodes, since episode k on a map uses the same random seed for each. A cell is the mean, over the three training seeds, of each seed's ten-map mean. Its *seed interval* is the lowest and highest of those three means, not a confidence interval. A teammate is *disbelieved* while the team's mean trust in it is below one half (for the reputation gate, while it is gated), and a liar is *detected* in an episode if it is disbelieved at least once.

6 Results

6.1 Exact sensing

Table 1 compares the methods by attacker family. Under exact sensing Momus wins 87.4% of episodes and the trained reputation gate 84.6%. Momus thus fails in 12.6% of episodes against 15.4%, nearly

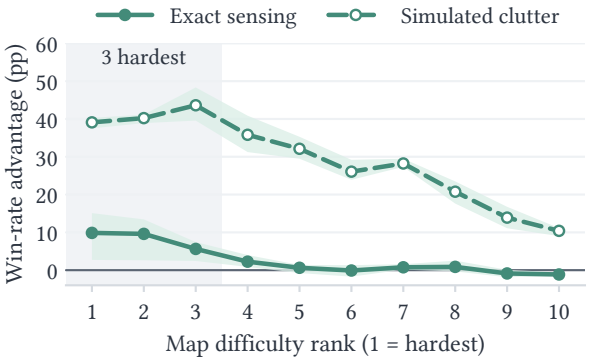


Figure 4: Momus's lead over the reputation gate is largest where pursuit is hardest. Win-rate advantage per evaluation map, averaged over the fourteen conditions. Maps are ranked by plain PPO's exact-sensing win rate. Points show the mean of three seed-matched differences and bands their range.

a fifth fewer failures or 2.8 points of win rate. Momus's worst training run beats the reputation gate's best in every summary row of supplementary Table S9 except the drop under attack. Averaged over seeds, Momus also leads every adapted defence in all fourteen conditions (supplementary Table S8). Momus generalises to attackers and maps unseen in training, winning 87.3% on held-out attackers and 87.8% on held-out maps against 86.5% and 86.9% on seen ones. The held-out attackers are not easier, since plain PPO wins 18 points less on them (49.5% against 67.8%).

The lead holds with no liar at all (93.9% against 90.2%), suggesting that its belief also locates the evader better. It is concentrated where the task is hard (Figure 4). The three maps with the lowest plain-PPO win rates are also the three densest. On them Momus wins 74.3% and the reputation gate 66.0%, and averaged over them Momus leads

Table 1: Win rate by attacker family under exact sensing. ∗: published defence adapted to this setting (§5). Family rows average maps, seeds and the family's conditions (three naive, two consistent, three rule-evading, four near-truth, one colluding). Summary rows give the 14-condition mean (without the learned attacker), the six held-out attackers and the three maps hardest for plain PPO. Brackets: seed interval (§5.3). †: held out of training. (†): some members held out. Bold: the smallest set of methods that each beat every method outside it on every matched seed, not a significance test.

Attacker family	Plain PPO	Reputation gate (MATE)* [19, 22]	Attention (AACFT)* [33]	Type head (BARDec)* [25]	AME* [37]	Momus
honest	0.880	0.902	0.874	0.876	0.854	**0.939**
naive fabrication†	0.175	**0.856**	0.180	0.170	0.247	**0.876**
consistent fabrication	0.558	0.836	0.586	0.598	0.626	**0.865**
rule-evading fabrication(†)	0.757	0.845	0.768	0.755	0.749	**0.866**
near-truth lies(†)	0.778	0.856	0.797	0.777	0.799	**0.893**
collusion	0.691	**0.745**	0.690	0.672	0.677	**0.764**
learned attacker	0.769	**0.858**	0.727	0.774	0.746	**0.868**
14-condition mean	0.614 [0.597–0.625]	0.846 [0.831–0.863]	0.626 [0.624–0.629]	0.616 [0.604–0.625]	0.640 [0.612–0.671]	**0.874** [0.873–0.874]
mean, held-out (6)	0.495 [0.488–0.505]	0.849 [0.836–0.865]	0.503 [0.502–0.507]	0.490 [0.486–0.495]	0.518 [0.490–0.547]	**0.873** [0.870–0.876]
mean, hardest maps (3)	0.426 [0.413–0.438]	0.660 [0.622–0.710]	0.467 [0.445–0.486]	0.444 [0.422–0.459]	0.449 [0.430–0.478]	**0.743** [0.735–0.757]

Table 2: Win rate by attacker family under simulated clutter, rows and markers defined as in Table 1. The other defences under the same clutter are in the supplement.

Attacker family	Plain PPO	Reputation gate (MATE)* [19, 22]	Momus
honest	0.634	0.599	**0.902**
naive fabrication†	0.348	0.549	**0.835**
consistent fabrication	0.481	0.557	**0.841**
rule-evading fabrication(†)	0.560	0.543	**0.844**
near-truth lies(†)	0.527	0.546	**0.842**
collusion	0.486	0.467	**0.712**
14-condition mean	0.494 [0.445–0.520]	0.546 [0.536–0.562]	**0.836** [0.830–0.840]
mean, hardest maps (3)	0.265 [0.239–0.281]	0.262 [0.253–0.267]	**0.672** [0.659–0.679]

by 4 to 12 points in every condition (supplementary Figure S1). On the other seven maps the two are within five points.

Why the learned defences trail. Plain PPO (61.4%) and the other adapted defences (61.6–64.0%) trail far behind. With no liar they win 85.4–87.6%, near plain PPO's 88.0%. So they learn the task but are misled by lies. None models what a sender could have observed, so an impossible claim looks like any other. The held-out decoy names a cell outside its own cone. Against it they win 12.7–18.2% and plain PPO 12.6%, while the reputation gate and Momus detect it at the first step. Attention and the type head only reweight or annotate reports, so their routing still follows the freshest claim. That claim is honest in three training episodes out of four. Their source papers target other threats, namely faults the environment flags (AACFT), corrupted actions (BARDec) and a bounded number of corrupted messages (AME). AME's certificate holds when enough of its subsets agree that one without the liar's report must have proposed the action. It held on 99.5% of steps against the learned attacker, yet AME lost a quarter of its episodes.

Collusion and the learned attacker. Two colluding liars give the lowest win rate for both Momus (76.4%) and the reputation gate (74.5%). Against the learned attacker Momus wins 86.8%, Momus without the verification target 86.1% and the reputation gate 85.8%, all within seed variation. The other defences win 72.7–77.4% (Table 1 and supplementary Table S11). Against Momus the attacker more often keeps the evader unsighted (2.1% against 0.9%), but the evader escapes less often (11.0% against 13.1%). One round of training may not find the strongest attack, so this is not a worst case.

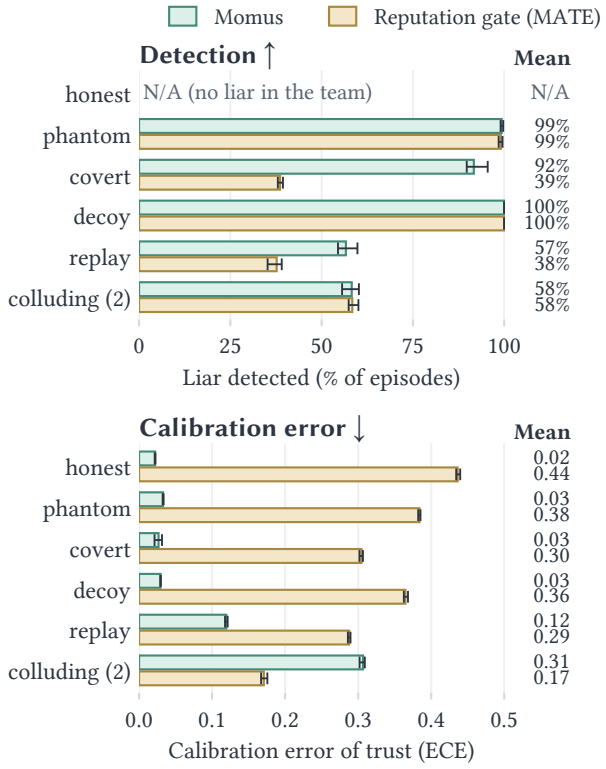


Figure 5: Momus detects plausible liars and calibrates its trust. Top: episodes in which a live liar is detected. Bottom: expected calibration error of trust against true type, taking the reputation gate's trust as its Beta mean. Bars show the seed mean (ECE pooled over seeds) and whiskers the seed range.

6.2 Simulated clutter

A false detection is broadcast like a true sighting. An honest teammate therefore soon names a cell that another teammate's cone covers and finds empty, or jumps between false cells faster than the evader could move. Either event counts against it. The reputation gate treats about half of honest reporter-steps as a liar's, against none under exact sensing (condition means 48.9–51.3%, supplementary Table S10). With no liar at all, its win rate falls below plain PPO's (reputation gate 59.9%, plain PPO 63.4%, Table 2). Across all conditions it wins 54.6%, 5.2 points above plain PPO's 49.4%. On the three hardest maps the two are level (26.2% and 26.5%).

Under clutter, judging a teammate means telling an honest false sighting from a lie. Through its positive ε floor (6), Momus's honest likelihood can explain an otherwise unsupported claim either way. The reputation gate instead counts every contradiction against the sender. In no condition does Momus disbelieve more than 2.3% of honest reporter-steps on average. It wins 83.6%, a relative failure-rate reduction of 63.9%. In every condition each of its three training runs beats each of the reputation gate's. Across the runs it wins more in 419 of 420 condition–map cells. On the hardest maps it wins 67.2%, more than twice the reputation gate's rate. Most of the gap is present with no liar at all (90.2% against 59.9%), where Momus also beats plain PPO by 27 points. The same judgement that keeps an honest teammate trusted also discounts its false sighting, which an undefended team follows. Liars still cost Momus 7.2 points on average and plain PPO 15.1.

6.3 Detection and calibration

Momus and the reputation gate agree where a lie is refutable (Figure 5). Both detect the decoy in every episode and the phantom in almost every one. They differ on covert, which is designed to evade instantaneous consistency checks. Momus detects it in 91.7% of episodes and the reputation gate in 38.6%. In the episodes each detects, the first detection comes on average at step 77 against 476. On replay Momus detects 56.6% and the reputation gate 37.7%. Against the colluding pair, Momus detects one of its liars in 58.2% of episodes and the reputation gate in 58.4%. Momus's trust is also better calibrated, with expected calibration error [17] 0.022–0.033 against the reputation gate's 0.305–0.436 on honest, phantom, covert and decoy. It also ranks the covert liar far better taken over all thresholds, with an area under the ROC curve (AUROC) [11] of 0.950 against 0.562, where 0.5 is chance (supplementary Table S3). On the colluding pair the reputation gate discriminates and calibrates better, likely because Momus's belief admits at most one liar (§6.5).

6.4 What the verification target adds

The ablation removes only the verification target, so the policy is no longer told which claim to check (supplementary Table S4). It still sees, for each belief mode, the trust in the nearest claim's sender and whether the mode is in view, and is still rewarded for sharpening trust (§5.3), so it can learn to check claims by itself, and it does. Against plausible liars (consistent and rule-evading fabrication), a pursuer's view already overlaps the envelope of the claim the gain rule would pick on 14.5% of the steps it holds one, against 13.4% with the target (supplementary Table S6). At six pursuers the two win rates are within seed variation, 87.4% with the target and 87.3%

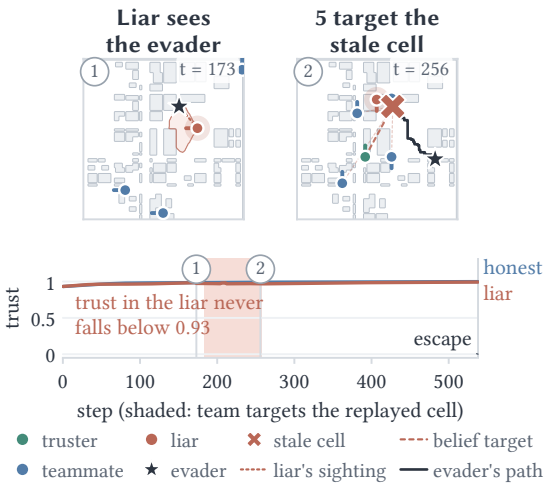


Figure 6: A failure: a replayed sighting keeps the liar trusted. Momus against the replay liar.

without. A likely reason is that six pursuers leave verification little to recover. The team finds the evader in 99.6% of episodes, and almost every loss is an escape during the chase (12.2 of 12.6 points).

The verification target points where a check can still change trust. Against plausible liars, a pursuer's target is the liar's claim on 63–69% of the steps it holds one, depending on the attacker. Naive liars are refuted at once, so their trust settles and their claims are the target on only 7–9% of such steps (supplementary Table S5). Near-truth liars, whose claims began as true sightings, are rarely the target (0–7%), which is the weakness discussed in §6.5.

Momus's policy relies on how its target is chosen. On the same trained policy, choosing the claim by information gain rather than at random raises the win rate on every seed, by 1.1 points on average, and loses in no tested condition (supplementary Table S7). Against choosing the newest claim it also gains on every seed, 0.8 points on average and almost all on the decoy (4.6 points). The decoy claims every step, so the newest-claim rule keeps pointing pursuers at a liar already exposed, which the gain rule rarely targets. We do not show that the target beats checks the policy learns by itself. Its value is that it is zero exactly when a check cannot inform (Proposition 4.2) and that it states which teammate to check and how much the check would reveal, so the choice can be audited.

6.5 Failure analysis

Momus loses most against the colluding pair (1,061 of 4,500 episodes). Collusion lies outside the one-liar belief, so the pair's agreeing reports read as corroboration, which a two-liar belief with a copying-liar model could address (§7). Figure 6 shows the weakness behind near-truth lies. A replayed sighting began as a real detection and moves no faster than the evader could. Neither the sender's cone nor the motion model contradicts it, so trust stays high while the team chases where the evader was. The liar stamps the replay with age zero, so catching it needs evidence about when a report was made, not only where.

7 Conclusion

Momus judges reports against what their senders could have observed while jointly locating the evader. Because it grades each report and each silence by its likelihood, it rarely mistakes an honest false detection for a lie, and under simulated clutter it has 64% fewer failures than the reputation gate. Under exact sensing it has nearly a fifth fewer failures, mostly on the densest maps, and it detects liars whose claims stay plausible. On the same policy, choosing targets by information gain beats random or newest-claim selection.

Future work can extend the belief to two liars, with a liar model in which one liar copies another's claim. This would bring a colluding pair inside the model, at the cost of sixteen type configurations per truster instead of six. Other directions are stronger adaptive attackers, transfer across map sizes, separate estimates of trust and location, a fully passive baseline, relaxing (A1)–(A4), decentralised execution and real sensors. We plan to release the benchmark and attacker code at a later date.

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